No license, no problems - But the data can't get too popular

Sharing the Renaissance Simulations

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Written by Amy Roberts and Jessica Walker from a conversation with John Wise on 13 October 2026, and approved by him.

The code has a license; the data never has

Enzo, an adaptive-mesh code for simulating the first stars and galaxies, has been open source since the late 1990s. It came out of Mike Norman's group, where Greg Bryan wrote the first version and where open-source astrophysics codes (Zeus before it) were already the norm; the source went out as tar files on FTP before there was a web to put it on. The first license was the NCSA open source license; Enzo is now BSD, hosted on GitHub. Matthew Turk pushed for taking licensing seriously; most of the developers, John included, did not much care who took the code or what they did with it, so long as it was open. And as a result of being open, it was very widely used.

The Renaissance Simulations data have no license at all. The data are available online by an opt-in process: people who download are asked to let the group know, and the group answers their questions. The goal is to support science, not preserve ownership to the data. And in fact we have developed a new collaboration based on our open data. The possible downside is that someone will interpret the simulations incorrectly, not that they will do anything bad with them.

Neither of the two public sites for the data states terms or asks for an acknowledgment. The older one, the Renaissance Simulations Laboratory, was built by Britton Smith about eight years ago to let users analyze the raw snapshots in hosted notebooks, with halo catalogs and merger trees alongside. However, the Renaissance Simulations Laboratory is now website-only because its key supporters have moved and retired. A newer site, the Renaissance Simulations Vault, built by an undergraduate and live since summer 2026, serves 25 GB of per-halo catalogs (masses, star-formation rates, stellar and gas properties for about 3,700 halos) rather than the raw snapshots.

One question a year, and that is about the limit

The group gets about one data question a year, which is manageable. If demand rose much above that, the handful of people who support the data could not keep up, because there is no funding for user support. Most of our time is spent helping more experienced users; one collaboration started that way, with a known colleague given Globus access who moved the data to the Ohio Supercomputer Center. And we sometimes get requests that we can't support: the first email through the new site asked for quantities the catalogs do not contain, and had to be turned down: computing them would take time the student project does not have.

Where the bytes live

The dataset is 30 TB. The raw data sit on ACCESS storage (Ranch, at UT Austin). The public copy is shared through Globus. ACCESS makes sharing possible; even hosting the 25 GB of catalogs locally drew objections from campus IT.

The pain point is sharing, not licensing

Each snapshot in time is 100–200 GB, so even one snapshot puts users into HPC territory. In practice the people who use the data already have resources somewhere and transfer the whole set to where they have space, which means the dataset is duplicated at every site that uses it. Ideally, the data would live in one place and be reachable from anywhere by anyone with permission, with analysis run in a notebook hosted where the data are, rather than copies moved to the compute. The Laboratory was a partial answer to this: not everyone wants to download all the data, and a notebook hosted next to it needs no download at all.

For the session

First, the absence of a data license here is not a decision; the opt-in request does the work a license and an acknowledgement clause would otherwise do. Second, access is limited less by terms than by logistics: whoever can move 30 TB can use the data, and whoever cannot, cannot, whatever the license says.

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